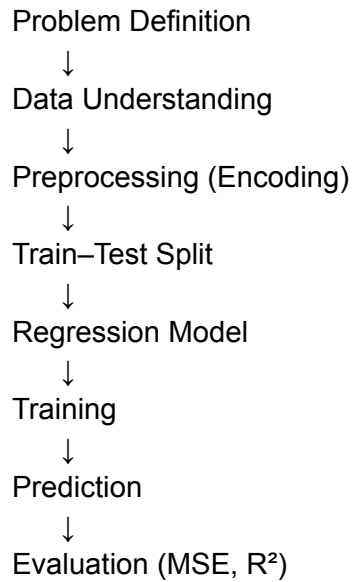


Pipeline



Problem Definition

Goal: Predict **house prices** using features such as:

- Location
- Size (area in sqft)
- Bedrooms, bathrooms
- Amenities (parking, balcony, furnished)
- Age of the house

This is a **regression task** because the output (price) is a **continuous numeric value**.

Step 1: Understand the Dataset

Each row represents one house.

Target variable (y):

- **Price** → house price (continuous)

Input features (X):

- Numerical: **Area_sqft**, **Bedrooms**, **Bathrooms**, **Age_Years**, **Parking**, **Balcony**
- Categorical: **Location** (Urban/Suburban/Rural), **Furnished** (Yes/No)

Step 2: Data Preprocessing

(a) Handle Categorical Variables

Machine learning models work with numbers, not text.

So we convert:

- **Location** → Urban / Suburban / Rural
- **Furnished** → Yes / No

We use **One-Hot Encoding**:

Location_Urban, **Location_Suburban**, **Furnished_Yes**

This avoids introducing false ordinal relationships.

(b) Feature–Target Split

We separate:

- **X** → all input features
- **y** → **Price**

This reflects the function we want to learn:

$$\text{Price} = f(\text{Location}, \text{Area}, \text{Bedrooms}, \text{Amenities}, \text{Age})$$

Step 3: Train–Test Split

We divide the data into:

- Training set (80%) → learn the relationship
- Test set (20%) → evaluate performance

Step 4: Choose the Regression Model

Linear Regression

Step 5: Train the Model

The model learns:

- How each feature contributes to price
- Coefficients (weights) that minimize prediction error

This is done by minimizing **Mean Squared Error (MSE)** on training data.

Step 6: Make Predictions

Step 7: Evaluate the Model

Metrics Used: MSE , R Square

Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum (y_{true} - y_{pred})^2$$

- Lower = better
- Penalizes large errors strongly